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(54) **RIGHTING DEVICE FOR FLOATING UNIT**

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(71) Applicant: **TOYOTA JIDOSHA KABUSHIKI KAISHA**, Toyota-shi (JP)

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(72) Inventors: **Tateru Fukagawa**, Fujisawa-shi (JP);
Mikikane Hirohata, Sunto-gun (JP);
Yoshiharu Saito, Gotemba-shi (JP);
Tetsuyuki Oe, Koganei-shi (JP);
Hayato Masuki, Fuji-shi (JP)

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(73) Assignee: **TOYOTA JIDOSHA KABUSHIKI KAISHA**, Toyota-shi (JP)

(57) **ABSTRACT**

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A righting device for righting a floating unit with a plurality of hulls from a capsize in a liquid includes a valve liquid injection unit capable of selectively performing a ballast function by injection of a liquid for each of the hulls, and a liquid displacement unit capable of selectively performing a float function by displacing the injected liquid for each of the hulls.

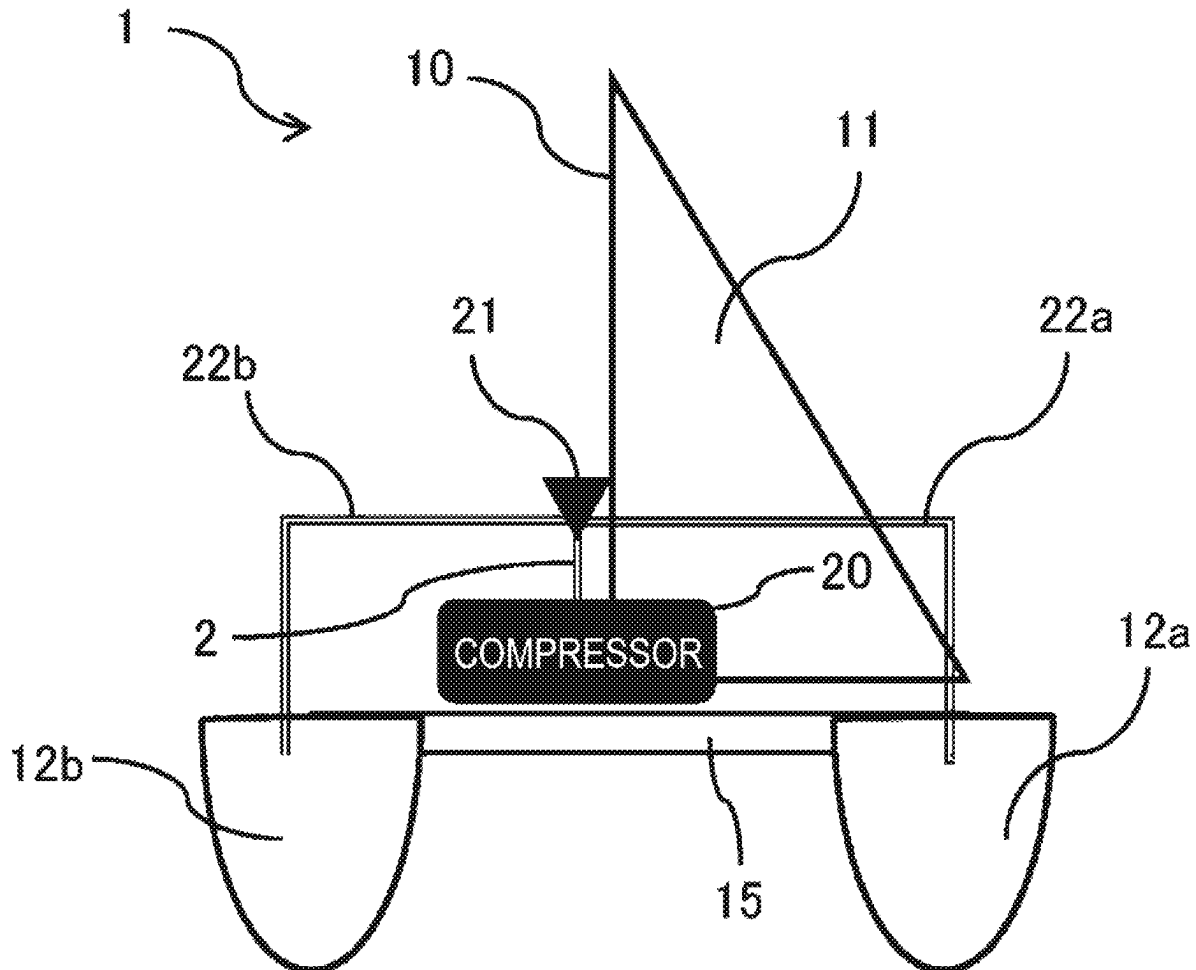


FIG. 1

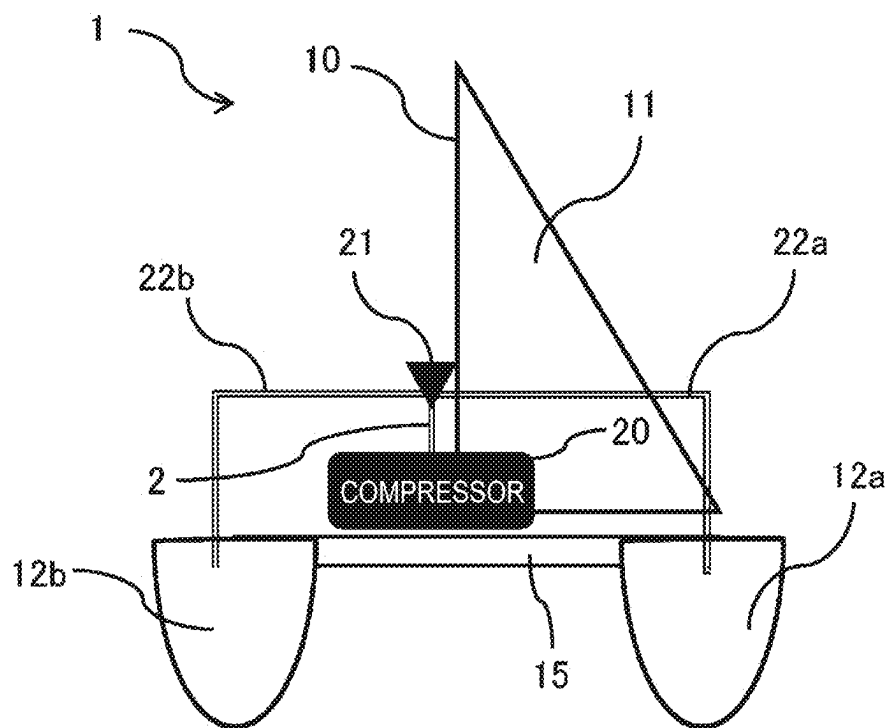


FIG. 2

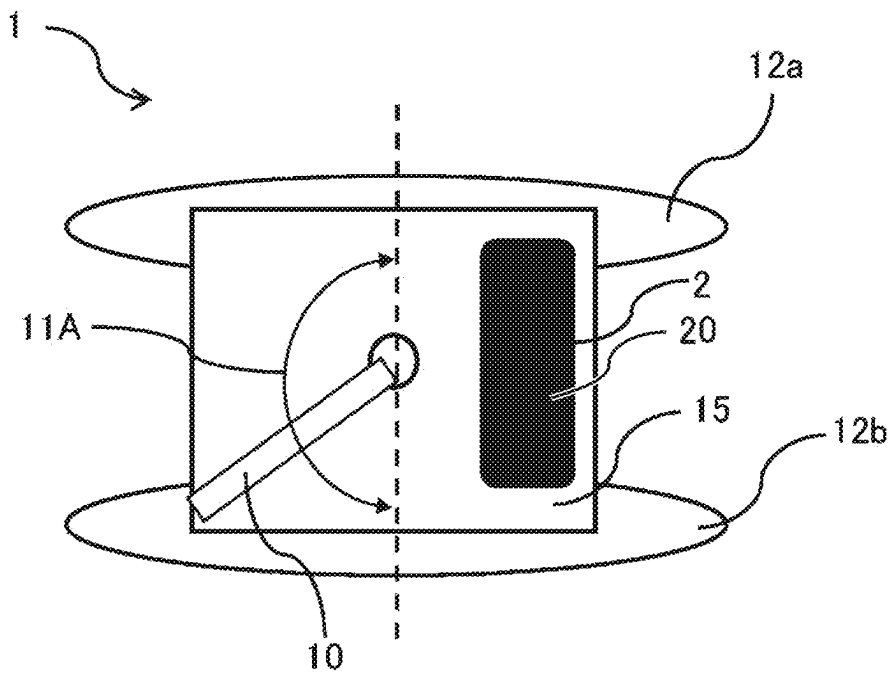


FIG. 3

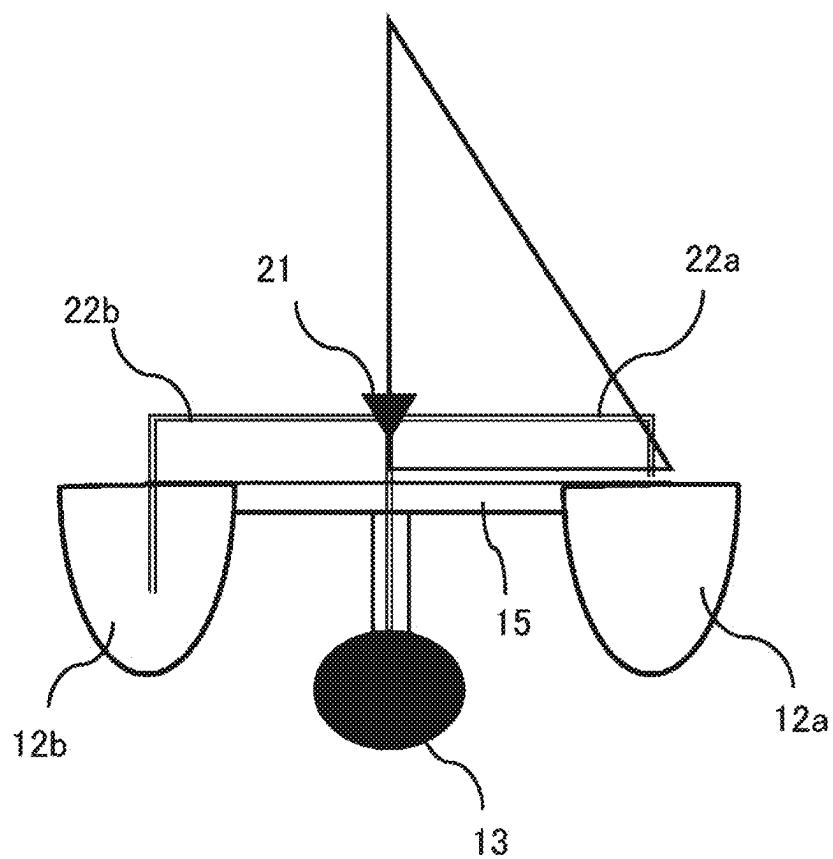


FIG. 4

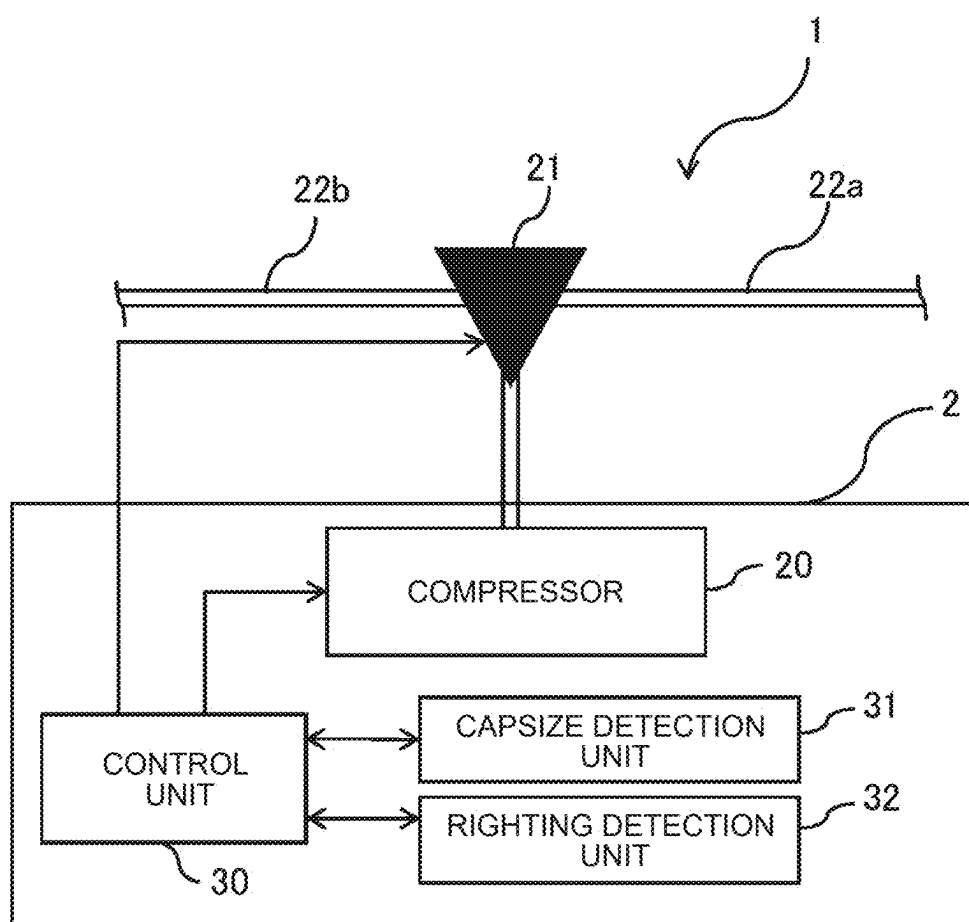
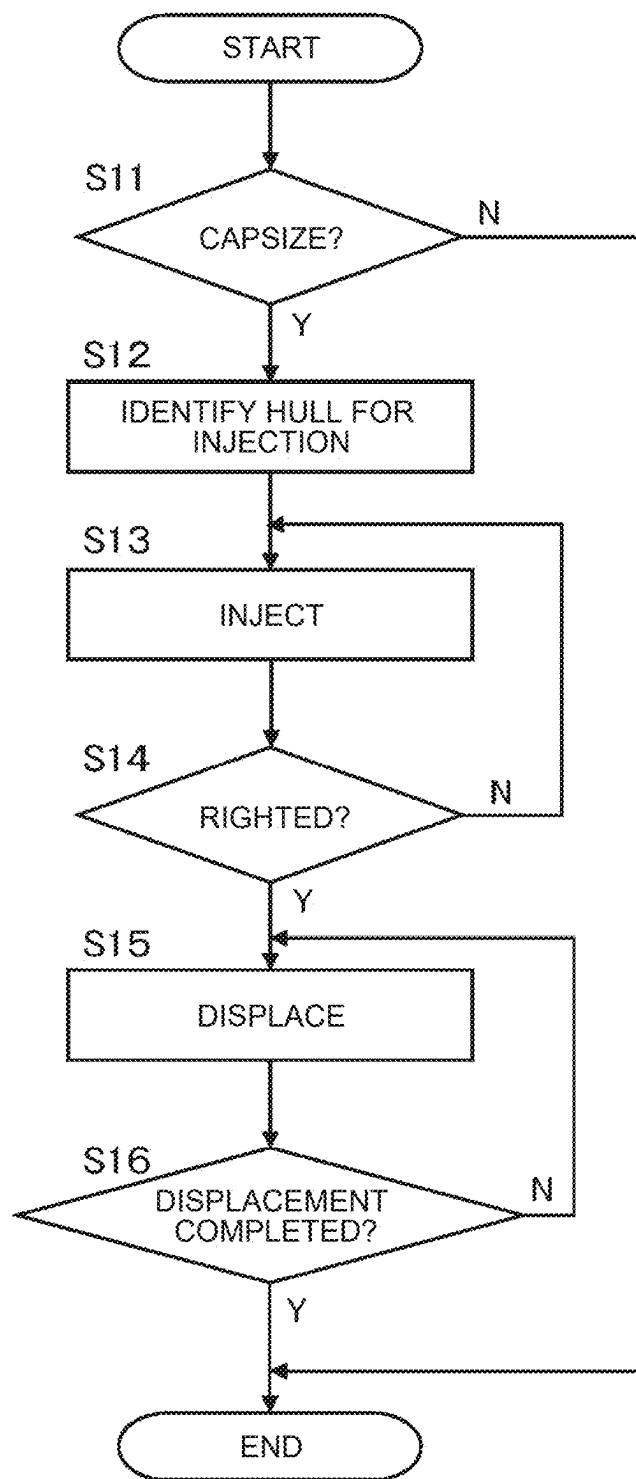


FIG. 5



RIGHTING DEVICE FOR FLOATING UNIT**CROSS-REFERENCE TO RELATED APPLICATION**

[0001] This application claims priority to Japanese Patent Application No. 2024-037366 filed on Mar. 11, 2024, incorporated herein by reference in its entirety.

BACKGROUND**1. Technical Field**

[0002] The present disclosure relates to righting devices that right a floating unit when the floating unit capsizes. The floating unit is a floating unit, such as a yacht, of systems using various types of kites, namely systems that fly a kite from a floating unit, such as a tethered wind power generation system, an observation system, a transportation system, and a communication system.

2. Description of Related Art

[0003] Various types of vessels with sails that fly a kite from their sterns have been proposed as this type of floating unit or an offshore unit. These types of vessels are expected not to easily capsize as a technique of stabilizing the attitude of a vessel, such as traditional ballast, or the existing multihull stabilization technique for vessels such as a catamaran having two hulls and a multihull having three or more hulls is applied to the vessels (see U.S. Pat. No. 10,029,773 B1).

SUMMARY

[0004] However, for example, a floating unit of a tethered wind power generation system is more likely to capsize than ordinary vessels. This is because such a floating unit is used in a special environment in which a kite is flown from the floating unit in order to use somewhat strong wind for power generation and, from the standpoint of energy efficiency, the power for moving the floating unit depends on the power of wind that hits the sails. Floating units of various systems using a kite, such as an observation system, a transportation system, and a communication system, are also likely to capsize depending on the magnitude of tension that is applied from the kite flying in a certain strength of wind to the floating unit via a tether. A multihull floating unit that is unrelated to kites may also capsize depending on the application or situation. The above related art has a technical issue. Namely, the above related art requires various actions to be taken in case of a capsize, such as getting the floating unit back afloat, sinking the floating unit, and towing the floating unit.

[0005] It is an object of the present disclosure to provide a righting device for a floating unit that can right a floating unit such as a yacht from a capsize.

[0006] An aspect of a righting device for a floating unit according to the present disclosure is a righting device for a floating unit that rights a floating unit with n hulls (where n is a natural number of 2 or more) from a capsize in a liquid. The righting device includes:

[0007] a liquid injection unit configured to selectively inject the liquid into the n hulls to selectively cause the n hulls to serve as ballast;

[0008] a liquid displacement unit configured to selectively displace the injected liquid from the n hulls to selectively cause the n hulls to serve as a float; and

[0009] a control unit configured to, in case of the capsize of the floating unit, control the liquid injection unit to inject the liquid into m hulls (where m is a natural number smaller than n) that are located at a relatively great depth in the liquid out of the n hulls, and when the floating unit is righted from the capsize by injection of the liquid, control the liquid displacement unit to displace the injected liquid from the m hulls.

[0010] According to the above aspect of the righting device for a floating unit of the present disclosure, when the multihull floating unit capsizes, the control unit first controls the liquid injection unit to inject the liquid into the m hulls (i.e., part of the hulls) of the floating unit. As a result, the m hulls serve as ballast, so that the floating unit flips back over. The control unit then controls the liquid displacement unit to displace the liquid from the m hulls. As a result, the n hulls (i.e., all of the hulls) serve as floats. The floating unit is thus restored to a properly functioning state before the capsize.

[0011] Such functions and effects of the present disclosure will become more apparent from an embodiment of the present disclosure described below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0013] FIG. 1 is a schematic side view showing an entire configuration of a righting device for a floating unit according to an embodiment together with a main part of the floating unit;

[0014] FIG. 2 is a schematic plan view of an embodiment;

[0015] FIG. 3 is a schematic side view illustrating an example of a main structure under the deck of the embodiment;

[0016] FIG. 4 is a block-diagram illustrating an exemplary main construction of a vessel of the embodiment; and

[0017] FIG. 5 is a flowchart illustrating an example of a righting process according to the embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0018] First, with reference to FIGS. 1 and 2, an overall configuration of a righting device for a floating unit according to an embodiment will be described. This embodiment is constructed, for example, in a multihull floating unit of a tethered wind power generation system that flies a kite from a floating unit such as a yacht or an offshore unit, as a righting device that rights the floating unit when it capsizes. Such a righting device can be applied to a multihull floating unit such as, for example, a catamaran having two unique hulls and a multihull having three or more hulls. In either case, righting can be performed relatively easily or quickly and can be performed automatically or semi-automatically without manual intervention, as will be described in more detail below.

[0019] As shown in FIGS. 1 and 2, the floating unit 1 of the multi-hull type includes a mast 10, a sail 11 (see FIG. 1) rotatable around the mast 10 at a movable range 11A (see FIG. 2), a hull 12a and a 12b, and a deck 15. It is constructed

as a floating unit of a tethered wind power generation system that skips a kite (not shown), and in particular, it includes a righting device 2 according to the present embodiment. The floating unit 1 is floated by the buoyancy of the hull 12a and 12b on the sea, the ocean, the lake, and the river, and is movable by the wind force received by the sail 11. Further, the floating unit 1 is configured to perform a movement suitable for wind power generation due to a change in tension received from a kite via a tether when a kite (not shown) is skipped, or to adopt a posture.

[0020] The righting device 2 includes a compressor 20 capable of selectively injecting air, which is an example of a gas, into a hull 12a and a 12b, a valve 21 capable of selectively injecting seawater, which is an example of a liquid, into the hull 12a and 12b, a pipe 22a and a 22b that selectively communicate between the valve 21 and the hull 12a and 12b, and a control unit (see FIG. 4), which will be described later.

[0021] The valve 21 is configured as a branch valve connected to the compressor 20 on the other side of the pipe 22a and 22b. By the operation of the compressor 20, the hull 12a and 12b are each selectively made to function as a float. The hull 12a and 12b are selectively caused to function as ballasts by injecting seawater through a flow path (not shown) separate from the compressor 20 via the valve 21. As described above, in the present embodiment, an example of the “liquid injection unit” is constituted by the valve 21 functioning as the seawater injection unit, and an example of the “liquid displacement unit” is constituted by the compressor 20 functioning as the gas injection unit or the air injection unit.

[0022] As shown in FIG. 3, a ballast 13 extending deep into the sea is provided below the deck 15 in the vicinity of a central position located between the hull 12a and 12b when viewed in a plan view. The floating unit 1 may be fixed by an anchor separately provided at a fixed position on the sea according to the situation. The floating unit 1 receives tension from a kite which is fixed to receive a certain degree or a maximum strong wind in one position for power generation, and tends to fall down or capsize during power generation. Furthermore, from the viewpoint of energy efficiency, the floating unit 1 is moved or tilted in an orientation different from the orientation of the tension from the kite by the wind force hitting the sail 11, so that the floating unit tends to fall or capsize during power generation. However, in the present embodiment, the presence of the ballast 13 makes it possible to reduce the risk of a capsize to some extent.

[0023] However, even with the ballast 13, in order to further increase the power generation efficiency, it is necessary to receive a stronger tension from the kite via the tether, which increases the risk of a capsize. Alternatively, if the risk of a capsize is reduced, the power generation efficiency of the kite will decrease, and the wind power of the folding angle will not be utilized. As a result, the ability to easily and quickly perform the righting process or the recovery process in case of a capsize is an extremely important factor in order to improve the power generation efficiency and recover the floating unit that has capsized. Therefore, in the present embodiment, as will be described in detail below with reference to FIGS. 4 and 5, a righting device 2 is constructed that can right the floating unit 1 quickly in a relatively simple manner from a capsize.

[0024] As illustrated in FIG. 4, in addition to the compressor 20 and the valve 21, the righting device 2 includes a control unit 30 that controls the on-off of the compressor 20 and the opening and closing of the valve 21 in the vessel of the floating unit 1, a capsize detection unit 31 that detects a capsize of the floating unit 1, and a capsize detection unit 31 that detects righting of the floating unit 1 from the capsize.

[0025] With reference to FIG. 5 in addition to FIG. 4, the configuration and operation of the righting device 2 will be further described. The operation processing illustrated in the flowchart of FIG. 4 is repeatedly executed by the control unit 30 as a subroutine periodically or irregularly during navigation or power generation of the floating unit 1 as described below.

[0026] That is, as shown in FIG. 5, first, the capsize detection unit 31 (see FIG. 4) determines whether the floating unit 1 has capsized during power generation or navigation of the floating unit 1 (S11). Specifically, whether the inclination of the floating unit 1 becomes 90 degrees or more is determined by the capsize detection unit 31 or by the cooperation work of the capsize detection unit 31 and the control unit 30, for example, by the angle sensor included in the capsize detection unit 31.

[0027] If it is determined that the floating unit has not capsized (S11: NO), the control unit 30 ends the series of processes.

[0028] On the other hand, if it is determined that the vehicle has capsized (S12: YES), the control unit 30 identifies a hull into which seawater is to be injected via the valve 21 (S12). Specifically, among the hull 12a and 12b, the hull located at a relatively great depth is identified as a hull into which seawater is to be injected.

[0029] Subsequently, under the control of the control unit 30, the injection of seawater into the hull specified by S12 is performed through the valve 21 and the pipe 22a or 22b corresponding to the hull (S13). When the seawater injection is performed in this way, the relatively downward position of the hull 12a and 12b selectively serves as a ballast, and the floating unit 1 that has capsized is rotated in a direction righting from the capsize, and finally is righted from the capsize.

[0030] Therefore, the righting detection unit 32 (see FIG. 4) determines whether the floating unit 1 has been righted from the capsize (S14). Specifically, for example, by the angle sensor included in the righting detection unit 32, it is determined whether the inclination of the floating unit 1 is righted within a predetermined inclination angle range assumed when the floating unit 1 is not capsized by the cooperation work of the righting detection unit 32 and the control unit 30 or by the righting detection unit 32.

[0031] Unless it is determined that the floating unit 1 has not been righted (S14: NO), seawater injection in S15 is continuously performed.

[0032] On the other hand, if it is determined that the floating unit 1 has been righted (S14: YES), the seawater displacement or seawater removal by the air injection is selectively performed with respect to the hull 12a or 12b injected with the seawater by S13 under the control of the control unit 30 via the compressor 20 and the pipe 22a or 22b corresponding to the hull (S15). When such seawater displacement is performed, the hull again serves as an original float, and the righted floating unit 1 is restored to a normal state before the capsize.

[0033] The state of the seawater displacement (S15) by the air injection of the compressor 20 is monitored by the control unit 30, and the seawater displacement is continuously performed (S15) until the seawater displacement is completed (S16: NO). Finally, when the seawater displacement is completed (S16: YES), the control unit 30 ends the series of processes. Thereafter, the floating unit 1 is navigated to a dock or towed for maintenance or repair. Alternatively, if there is no abnormality, failure, damage, or the like due to a capsize, the tether-type wind power generation is performed again.

[0034] As described above, according to the righting device 2 of the present embodiment, when the multihull floating unit 1 capsizes in the sea, seawater is injected into one hull (that is, hull 12a or 12b) of the floating unit 1 by the valve 21 under the control of the control unit 30. Then, the hull serves as a ballast, and the floating unit 1 is reversed in its upper and lower directions. Furthermore, under the control of the control unit 30, air injection is performed on the hull by the compressor. Then, all the hulls (i.e., the hull 12a and 12b) serve as floats, and the floating unit 1 is restored to a properly functioning state before the capsize. By adopting the control by the control unit 30 provided in the vessel in this way, in addition, by using the detection result by the capsize detection unit 31 and the righting detection unit 32, it is possible to execute the righting process automatically or semi-automatically without the help, the present embodiment is very convenient in practical use.

Appendix

[0035] With regard to the embodiments described above, the following additional notes are further disclosed.

(Appendix 1)

[0036] A righting device for a floating unit of Appendix 1 according to the present disclosure is a righting device that rights a floating unit with n hulls where n is a natural number of 2 or more from a capsize in a liquid, the righting device comprising: a liquid injection unit configured to selectively inject the liquid into the n hulls to selectively cause the n hulls to serve as ballast; a liquid displacement unit configured to selectively displace the injected water from the n hulls to selectively cause the n hulls to serve as a float; and a control unit configured to, in case of the capsize of the floating unit, control the liquid injection unit to inject the liquid into m hulls where m is a natural number smaller than n that are located at a relatively great depth in the liquid out of the n hulls, and when the floating unit is righted from the capsize by injection of the liquid, control the liquid displacement unit to displace the injected liquid from the m hulls.

[0037] According to the righting device of the floating unit described in Appendix 1, when a multihull floating unit such as a catamaran or a multihull capsizes due to an operation error or error of a kite in a case where the floating unit is applied to wind and rain, a wave, a load collapse, a failure, a damage, a collision, tethered wind power generation, or the like in a sea, a lake, a river, or the like, liquid injection of seawater, lake water, river water, or the like is performed for m (that is, a part) hulls of the floating unit by a liquid injection unit such as a valve under control of a control unit. These m hulls then serve as ballasts, and the floating unit is

reversed again. Further, under the control of the control unit, the liquid displacement unit such as a compressor, a gas generator, and a gas injector, for these m hulls, air injection, gas generation, gas injection, or the like of the liquid by gas injection such as hydrogen injection is performed. Then, n (i.e., all) hulls serve as floats, and the floating unit is restored to a properly functioning state before the capsize.

[0038] The above “liquid injection unit” is not limited to a “valve” such as a mechanically driven valve, a motor-operated valve, or a solenoid valve, and may include a lid, a hatch, a screw, an actuator, a pump, or the like that can inject liquid into the hull by mechanical or electrical or electromagnetic driving.

[0039] The above-described “liquid displacement unit” is not limited to a “compressor”, and may include a gas generator (in other words, a device capable of discharging a liquid) capable of injecting a gas into a hull (in other words, a device that functions only once and generates a gas in a manner such as an on-vehicle airbag), a device that allows a hydrogen gas generated on a floating unit to flow in when applied to a tether-type power generation system, a suction pump, an actuator, and the like.

(Appendix 2)

[0040] The righting device for a floating unit according to Appendix 2 according to the present disclosure is the righting device for a floating unit according to Appendix 1 that further includes a capsize detection unit that detects a capsize of the floating unit, and a righting detection unit that detects righting of the floating unit from the capsize, wherein the control unit controls the liquid injection unit so as to inject the liquid when the capsize is detected by the capsize detection unit, and controls the gas injection unit so as to inject the gas when the righting detecting unit detects the righting.

[0041] According to the righting device of the floating unit according to Appendix 2 of the present disclosure, when the floating unit capsizes, for example, the capsize is detected by a capsize detection unit such as a roll angle sensor, a G sensor, a water pressure sensor at an upper portion of the sail, and a capsize determination unit. Typically, the capsize is detected when the floating unit is tilted to 90 degrees or more. Then, under the control of the control unit, liquid injection such as seawater injection is performed for m hulls (i.e., a part) located relatively downward in the floating unit. As a result, m hulls serve as ballasts, and the floating unit is reversed in its upper and lower directions. That is, it is righted from the capsize. Such righting is detected by, for example, a righting detection unit such as a roll angle sensor, a G sensor, a water pressure sensor at an upper portion of the sail, or a righting determination unit. Typically, the inclination of the floating unit is detected as being within the range of the inclination angle corresponding to the non-capsizing state. Then, under the control of the control unit, for these m hulls, liquid displacement such as air injection by operation of the compressor is performed in this time. As a result, the n hulls serve as the original float, and the floating unit is restored to a properly functioning state before the capsize.

(Appendix 3)

[0042] According to appendix 3 of the present disclosure, there is provided a righting device for a floating unit, wherein the floating unit is configured as a unit that floats on

the liquid in a tethered wind power generation system having a kite and a tether whose distal end is attached to the kite and whose proximal end is attached to the floating unit.

[0043] According to the righting device of the floating unit according to Appendix 3 of the present disclosure, in particular in the case of the floating unit constituting the tether-type buoyancy power generation system, due to the nature of the wind power generation by utilizing the tension of the kite applied to the floating unit via the tether, since it tends to capsize, the effect of the righting device is remarkably or maximally exhibited. However, not only in the case of tether-type wind power generation, but also in the floating unit constituting a system that employs a kite for some purpose such as observation, transportation, and communication, the effect of the righting device is exhibited in accordance with the magnitude of tension received from the kite via the tether and the ease of capsizing of the floating unit in accordance with the wind force received by the sail during navigation or operation. Further, even when there is no kite, by applying the righting device, the effect can be exhibited in accordance with the ease of capsizing the floating unit.

(Appendix 4)

[0044] The floating unit according to Appendix 4 of the present disclosure includes the fall prevention device according to any one of Appendices 1 to 3.

[0045] According to the floating unit described in Appendix 4 of the present disclosure, even when the multihull floating unit capsizes, the righting device described in any one of Appendices 1 to 3 described above can restore the floating unit to a properly functioning state before the capsize.

[0046] The present disclosure can be modified as appropriate within the scope and spirit of the disclosure that can be read from the claims and the specification as a whole, and a righting device of a floating unit or a floating unit accom-

panied by such a modification is also included in the technical idea of the present disclosure.

What is claimed is:

1. A righting device for a floating unit that rights a floating unit with n hulls (where n is a natural number of 2 or more) from a capsize in a liquid, the righting device comprising:

a liquid injection unit configured to selectively inject the liquid into the n hulls to selectively cause the n hulls to serve as ballast;

a liquid displacement unit configured to selectively displace the injected liquid from the n hulls to selectively cause the n hulls to serve as a float; and

a control unit configured to, in case of the capsize of the floating unit, control the liquid injection unit to inject the liquid into m hulls (where m is a natural number smaller than n) that are located at a relatively great depth in the liquid out of the n hulls, and when the floating unit is righted from the capsize by injection of the liquid, control the liquid displacement unit to displace the injected liquid from the m hulls.

2. The righting device according to claim 1, further comprising:

a capsize detection unit configured to detect the capsize of the floating unit; and

a righting detection unit configured to detect righting of the floating unit from the capsize, wherein the control unit controls the liquid injection unit to inject the liquid in response to detection of the capsize by the capsize detection unit, and controls a gas injection unit to inject a gas in response to detection of the righting by the righting detection unit.

3. The righting device according to claim 1, wherein the floating unit is configured as a unit of a tethered wind power generation system that floats on the liquid, the tethered wind power generation system including a kite and a tether whose distal end is attached to the kite and whose proximal end is attached to the floating unit.

4. A floating unit comprising the righting device according to claim 1.

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